Midterm 1 Practice Problems 2

This is very roughly the level and quantity of problems you can expect on the midterm.

Practice 2.1

For each statement in the list below, explain whether the statement is true or false, and prove your answer.

Wherever the symbol X appears, it denotes a metric space. Unless otherwise stated, assume the usual distance function on $\mathbb{R}$.

(1) Every nonempty subset of $\mathbb{Q}$ which is bounded below in $\mathbb{Q}$ has a greatest lower bound in $\mathbb{R}$.

is also bounded.

(2) Given $A, B \subset \mathbb{R}$, let

$$A \cdot B = \{xy \in \mathbb{R} \mid x \in A, y \in B\}.$$

If A and B are bounded above, then $A \cdot B$ is bounded above.

(3) The ordered set $\mathbb{N}$ has the least upper bound property.

(4) The set of all functions $f : \mathbb{N} \to \{1, 2, 3\}$ is countable.

(5) If $E \subset \mathbb{R}$ is finite, then E is closed.

(6) If $E \subset X$, and E° is open in X, then E is open in X.

(7) Given any collection of open intervals in $\mathbb{R}$, the intersection of the collection is open.

(8) If every point not in $E \subset X$ is a limit point of E, then E is dense.

Practice 2.2

Show that every ordered field is infinite.

Practice 2.3

Let X be a metric space and $E \subset X$. Define E' to be the set of all limit points of E. Prove that E' is closed.

Practice 2.4

Show that the intersection of a family of compact sets $\{K_\alpha\}$ of a metric space X is compact.

Practice 2.5

Let X be a metric space, and let G be an open subset of X. Show that there exists a collection of open neighborhoods in X whose union is G.